

Observed Variable Models Model Fit

Agenda

- What makes a model “good”?
- Covariance Reproduction
- Degrees of Freedom in SEM
- Chi-square test of model fit
- Model fit indices
- Alternative methods for thinking about model fit

What do you think a “good” model is/does?

The Goal of Evaluating Models

- A model is a simplification of reality which still helps us understand reality
 - We want to simplify but not over simplify
 - Balancing simplification and loss of information
- Is the degree of misfit acceptable?

A Caveat

- All models are wrong, meaning that no model is correct
 - Correct models are not models
- Evaluating model fit is more art than science
 - Blind adherence to rules of thumb is harmful to our field
 - Rules of thumb can be helpful in getting you started with some intuition about relative performance
- Just because a model “fits well” does not mean it’s a good model
 - Poorly fitting models are probably not good models
- There are many ways to evaluate “model fit” and generally we look for agreement among methods, but methods will not always agree
- By the end of this lecture you WILL NOT be able to perfectly determine whether any model “fits well”

Explained Variance

Typically we evaluate regression models based on explained variance in the outcome: R^2

In an SEM we have multiple endogenous variables, and all variance is explained because the deviances are inherently part of the model.

We could calculate variance explained for different endogenous variables, but this is not aligned with how we think about “model fit” for SEM

Covariance Reproduction

Much of SEM evaluation focuses on the ability of the model to reproduce the sample covariance matrix.

Remember an assumption of SEM is that the population covariance matrix equals the model implied covariance matrix given population parameters

$$\Sigma = \Sigma(\theta)$$

Neither Σ or $\Sigma(\theta)$ are known, and so we work with their estimates

$$S \approx \Sigma(\hat{\theta})$$

When these two matrices are *very close* we say the model “fits well”

The Residual Matrix

There is a matrix calculated in SEM called the **residual matrix**

It is the differences between the sample covariance matrix and the estimated model implied covariance matrix

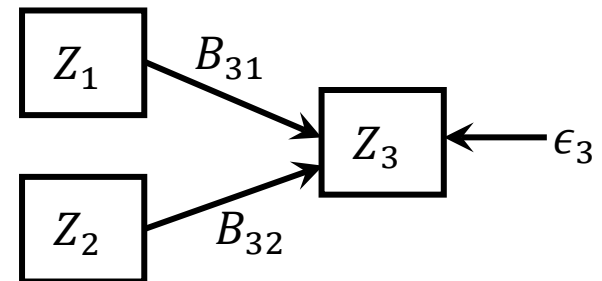
$$R = S - \Sigma(\hat{\theta})$$

Consider the following example:

$$S = \begin{bmatrix} 1 & .2 & .5 \\ .2 & 2 & .4 \\ .5 & .4 & 3 \end{bmatrix} \quad \Sigma(\hat{\theta}) = \begin{bmatrix} 1 & 0 & .469 \\ 0 & 2 & .306 \\ .469 & .306 & 2.971 \end{bmatrix}$$

$$R = \begin{bmatrix} 0 & .2 & .031 \\ .2 & 0 & .094 \\ .031 & .094 & .029 \end{bmatrix}$$

Are these values big or small?
Hard to tell because they are in covariance units



Standardized Residual Matrix

We can calculate the residual matrix with respect to the implied correlation matrix (i.e., standardizing all observed variables) and the residuals are in a more interpretable unit.

There are a few options for how to standardize, we will use the method proposed by Bentler (1995).

Both the observed and model implied covariance matrices are rescaled by dividing the elements by the square roots of the corresponding variances of the observed covariance matrix.

All entries are constrained to values between -1 and 1

$$\mathbf{R}^* = \begin{bmatrix} 0 & .141 & .018 \\ .141 & 0 & .038 \\ .018 & .038 & 0.010 \end{bmatrix}$$

Calculating Residuals in Lavaan

```
library(lavaan)

MRcov <- matrix(data = c(1, .2, .5, .2, 2, .4, .5, .4,
3), nrow = 3, ncol = 3, byrow = TRUE)
rownames(MRcov) <- c("Z1", "Z2", "Z3")
```

MRcov will be our observed covariance matrix

$$\mathbf{S} = \begin{bmatrix} 1 & .2 & .5 \\ .2 & 2 & .4 \\ .5 & .4 & 3 \end{bmatrix}$$

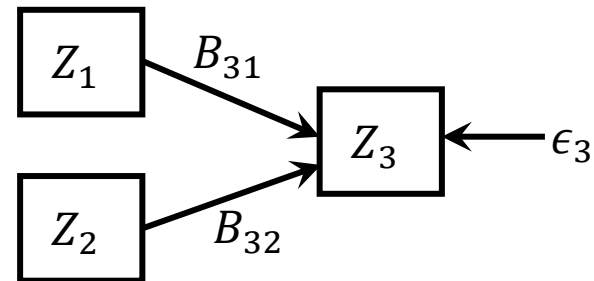
Because `lavaan` needs the variable names we provide them as row (or column) names as part of the matrix

Calculating Residuals in Lavaan

```
MRmodel <- 'Z3~Z1+Z2; Z1~~0*Z2'
```

```
MRfit <- sem(MRmodel, sample.cov = MRcov, sample.nobs =  
50, sample.cov.rescale = FALSE)
```

We specify the model:



And fit it using the `sem()` function. Notice I have to provide a sample size (`sample.nobs`) because that information is not in the covariance matrix.

I ran into a funny problem, which led me to discover something new about lavaan

The `sample.cov.rescale` argument

If the estimator is ML (the default), then the sample variance-covariance matrix will be rescaled by a factor $(N-1)/N$. The reasoning is the following: the elements in a sample variance-covariance matrix have (usually) been divided by $N-1$. But the (normal-based) ML estimator would divide the elements by N . Therefore, we need to rescale. If you don't want this to happen (for example in a simulation study), you can provide the argument `sample.cov.rescale = FALSE`.

Calculating Residuals in Lavaan

#gives the model implied covariance matrix
`fitted(MRfit)`

#gives the raw residual covariance matrix
`resid(MRfit)`

#gives the standardized residual covariance matrix
`lavResiduals(MRfit)`

$$\Sigma(\hat{\theta}) = \begin{bmatrix} 1 & 0 & .469 \\ 0 & 2 & .306 \\ .469 & .306 & 2.971 \end{bmatrix}$$

$$\mathbf{R} = \begin{bmatrix} 0 & .2 & .031 \\ .2 & 0 & .094 \\ .031 & .094 & .029 \end{bmatrix}$$

$$\mathbf{R}^* = \begin{bmatrix} 0 & .141 & .018 \\ .141 & 0 & .038 \\ .018 & .038 & 0.010 \end{bmatrix}$$

Just Identified Models

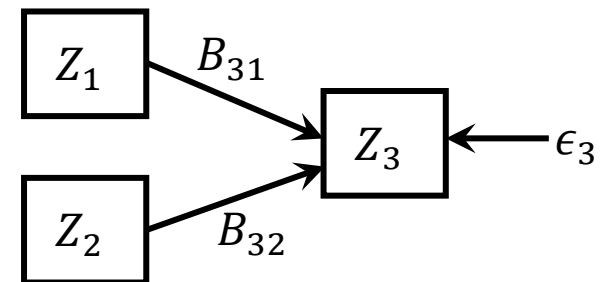
Just Identified Models are models where there are exactly as many parameters as there are variance/covariances available

$$\theta = \frac{1}{2} (N_Z)(N_Z + 1)$$

Just identified models have a model implied covariance which perfectly matches the covariance matrix

Consider our example:

We have $\frac{1}{2} (3)(4) = 6$ available pieces of information



The model estimates 5 parameters: $\sigma_1^2, \sigma_2^2, \sigma_3^2, B_{31}, B_{32}$

If we added one more parameter σ_{12} the model would be just identified. The implied covariance would exactly match the sample covariance.

Just Identified Model with Lavaan

```
#Just Identified Models  
MRmodelJI <- 'Z3~Z1+Z2'  
MRfitJI <- sem(MRmodelJI, sample.cov = MRcov, sample.nobs = 50,  
sample.cov.rescale = FALSE)  
summary(MRfitJI)  
resid(MRfitJI)
```

$$\mathbf{R} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Just identified models exactly recreate the sample covariance matrix.

This is not because they are “good models” but because they are not “models” they do not simplify the information.

Degrees of Freedom

Just identified models exactly recreate the sample covariance matrix.

This is not because they are “good models” but because they are not “models” they do not simplify the information.

We want models which simplify the information, and so we should prefer models with fewer parameters.

We can quantify the degree of simplification using **degrees of freedom**

$$df = \frac{1}{2} (N_Z)(N_Z + 1) - N_\theta$$

Degrees of freedom is the difference between the available information from the sample covariance matrix, and N_θ is the number of parameters

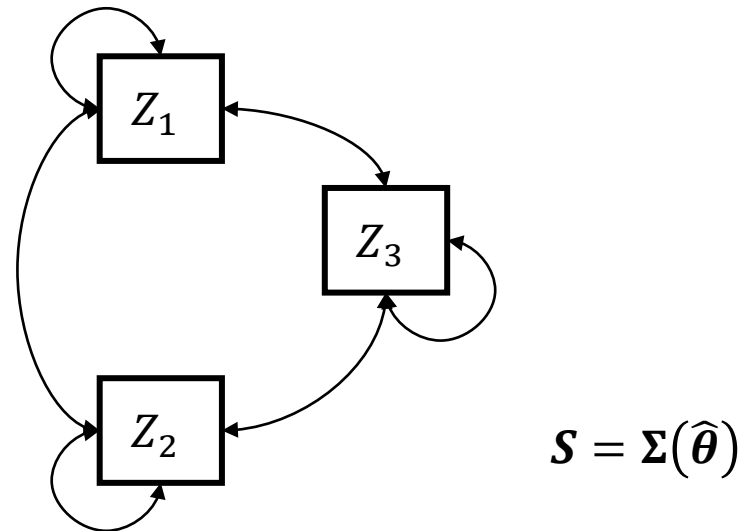
More degrees of freedom = more simplification (i.e., fewer parameters)

Fun fact: Multiple regression models have 0 degrees of freedom (they are just identified), this model “fits perfectly” in a trivial way

Saturated Models

Saturated models are a specific just identified model which estimate all possible variances and covariances.

A saturated model makes no restrictions on the covariance structure. It fits perfectly but has no degrees of freedom.



We can use this saturated model as a point of comparison for new proposed models.

If we can find models which fit similarly well, but impose new restrictions, those would be good models

Chi-square Model Fit

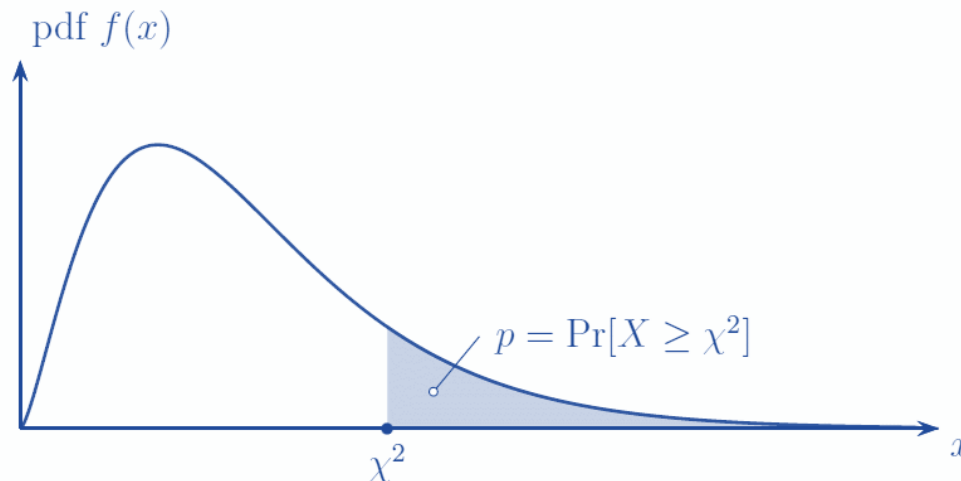
Under the Null Hypothesis of perfect model fit...

$$\mathbf{\Sigma} = \mathbf{\Sigma}(\boldsymbol{\theta})$$

We know the distribution of the fit function...

$$(N - 1)F_{ML} \sim \chi^2 \left(\frac{1}{2} (N_Z)(N_Z + 1) - N_{\theta} \right)$$

Given a χ^2 statistic, we can calculate a p -value which is the probability of observing a χ^2 equal to or larger than the observed value



Chi-square Model Fit

The null hypothesis is that the population covariance matrix is equal to the model implied covariance matrix.

Rejecting H_0 means there is **evidence of misfit** of our model with the population

When p is small: there is sufficient evidence that the null hypothesis is not true

When p is large: there is not sufficient evidence that the null hypothesis is not true

Be wary of accepting the null!

```
summary(MRfit)
```

```
Model Test User Model:
```

Test statistic	1.010
Degrees of freedom	1
P-value (Chi-square)	0.315

χ^2 test

Assumptions

- The observed variables have a multivariate normal distribution
- N is sufficiently large (so that the maximum likelihood properties kick in)
- None of the parameters are at their boundary (e.g., $\sigma_1^2 = 0$)

Limitations

- This is a formal hypothesis test, not really a holistic evaluation. We already know the null is false, we want to know how far we can deviate from the null before our model loses fidelity
- The logic of the test is reversed from our usual method (we want to fail to reject the null hypothesis)

This is still an important statistic, and you should always report it. However, rejection of H_0 does not necessarily mean your model is unacceptable

Model Fit Indices

There are two main types of fit indices:

1. **Incremental/Relative Fit Indices**

Compare your model to another typically “bad” model

We’ll discuss: CFI/TLI

Others: NFI, IFI, RNI

2. **Absolute Fit Indices**

No comparison to another specific model

We’ll discuss: SRMR, RMSEA

Others: GFI (and all subsidiaries), RMR

Standardized Root Mean Square Residual (SRMR)

SRMR relies on the residual matrix described earlier R

Joreskog and Sorbom (1981) proposed the root mean square residual, where we...

- take all the elements of R
- square them
- sum them
- divide by the number of parameters
- take the square root

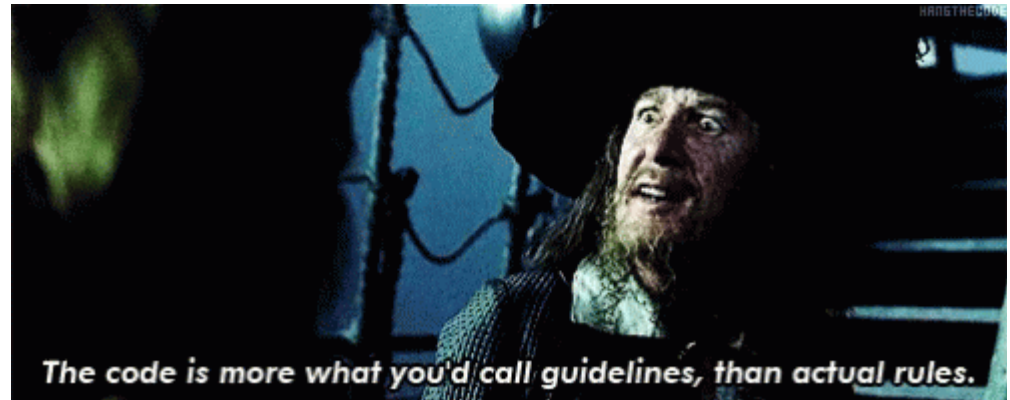
r_i

SRMR is the same but uses the standardized residuals instead (Bentler, 1995)

This allows comparison across models of different datasets

SRMR

- SRMR is a measure of “badness of fit” (higher values mean worse fit)
- $SRMR > 0$ (should be less than 1)
- Absolute fit index
- Proposed cutoff is .08



Limitations

- No penalty for model complexity
- Sensitive to sample size

RMSEA

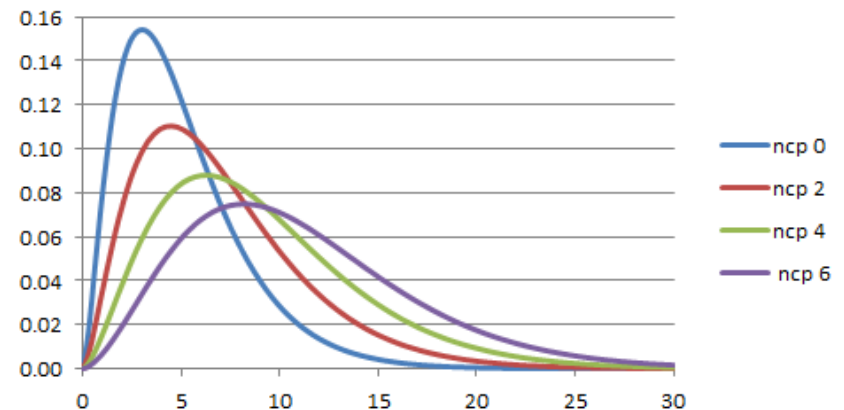
Measures discrepancy between observed and estimated covariance matrix, scaled by degrees of freedom

When the null hypothesis of “perfect fit” is violated, the χ^2 statistic has a non-central χ^2 distribution, defined by a non-centrality parameter

The noncentrality parameter affects center and spread of distribution

$(\chi^2 - df)/(N - 1)$ is the estimated noncentrality parameter for the χ^2 distribution if the null hypothesis is false

Noncentral Chi-square (df = 5)



$$\widehat{RMSEA} = \sqrt{\frac{\max(\chi^2 - df, 0)}{df(N - 1)}}$$

Many computer programs provide a 90% CI or a “test of close fit” where the null hypothesis is $RMSEA \leq .05$

RMSEA

RMSEA is a measure of **badness of fit** (higher numbers mean worse fit)

Values are greater than 0 (no upper limit)

Absolute fit index (no comparison to other models)

Penalizes for complexity by dividing by df

Cut-off recommendations: $< .05$ for “close fit”, $< .08$ for “reasonable fit” (Browne & Cudeck, 1992; Hu & Bentler, 1999)

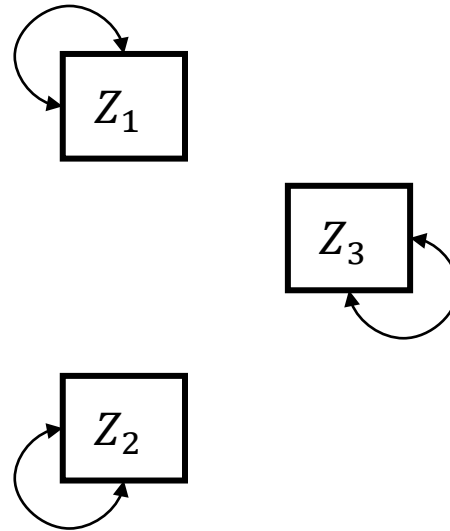


Limitations:

- Sensitive to sample size (underestimates fit at small samples $N < 200$)

Baseline Model

All variables get a variance, no variables covary with any other variables



This is a model with likely “too many restrictions” and we can compare our model to this model

CFI/TLI

CFI and TLI rely on a similar logic to RMSEA, but use a model comparison approach. They both compare to the **baseline model**.

Both rely on the intuition that under perfect fit $E(\chi^2) = df$

CFI uses a difference approach:

$$\frac{\max(\chi_0^2 - df_0, 0) - \max(\chi_k^2 - df_k, 0)}{\max(\chi_0^2 - df_0, 0)}$$

How much further off of the degrees of freedom is the proposed model k compared to the baseline model

TLI uses a ratio approach:

$$\frac{\chi_0^2/df_0 - \chi_k^2/df_k}{\frac{\chi_0^2}{df_0} - 1}$$

CFI/TLI

Both are measures of **goodness of fit**

Both should provide values between 0 and 1 (TLI can be negative)

Incremental/relative fit indices (relative to baseline model)

Penalize for complexity using degrees of freedom

Cut-off recommendations: $> .90$ for “good fit”, $> .95$ for “excellent fit” (Hu & Bentler, 1998)



Benefit:

- Not affected by sample size*

Requesting Fit Indices in Lavaan

```
summary(MRfit, fit.measures = TRUE)
```

User Model versus Baseline Model:

Comparative Fit Index (CFI)	0.997
Tucker-Lewis Index (TLI)	0.991

Loglikelihood and Information Criteria:

Loglikelihood user model (H0)	-255.039
Loglikelihood unrestricted model (H1)	-254.533
Akaike (AIC)	520.077
Bayesian (BIC)	529.637
Sample-size adjusted Bayesian (BIC)	513.943

Root Mean Square Error of Approximation:

RMSEA	0.014
90 Percent confidence interval - lower	0.000
90 Percent confidence interval - upper	0.375
P-value RMSEA <= 0.05	0.344

Standardized Root Mean Square Residual:

SRMR	0.060
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Criticisms of Fit Indices

- Cut offs are generated by studies of specific models (typically two factor CFA models) and may not generalize
- Different fit indices are sensitive to different types of misspecifications
 - This can lead to disagreement between fit indices
 - Hu and Bentler (1999) recommend seeking agreement between two (prespecified) fit indices (e.g., RMSEA and CFI)
- Fit indices are sensitive to
 - Strength of different model parameters (e.g., factor loadings)
 - Number of variables (latent and observed)
 - Sample size

More Recent Extensions of Fit Indices

- Equivalence testing approach to fit indices (Yuan et al., 2016)
 - A “high” value for CFI does not necessarily mean that we have evidence of good fit (it could be sampling variability). This method accounts for sampling variability and tests against a null hypothesis that the fit index is higher/lower than traditional cut-off recommendations
- Dynamic Fit Indices (McNeish & Wolf, 2021)
 - The appropriate cutoffs for fit indices differ based on the model. DFI takes a simulation approach to determine the appropriate cutoffs for different fit indices based on the model structure.

Alternative Ways to Think about Fit

Cross validation (Cudeck & Browne, 1983)

Split sample and test ability to reproduce out of sample covariance matrix (can use k-fold cross validation)

Wysocki et al. (2022) [cvsem](#) package in R

Local fit

Evaluate fit in different parts of the model

More on this when we get to more complex models

Information Criteria

AIC & BIC can be used to select amongst non-nested models

Sensitivity analyses

Do choices within my model affect the conclusion that I make about my primary parameters?

Model Fit is Orthogonal to Path Significance

It is possible to have a model which fits well, where no paths are significant

It is possible to have a model which fits poorly, and every path is significant

A little discussion



References

- Bentler P. M. (1995). *EQS 5* [Computer program]. Multivariate Software Inc.
- Browne, M. W., & Cudeck, R. (1992). Alternative ways of assessing model fit. *Sociological Methods & Research*, 21(2), 230-258.
<https://doi.org/10.1177/0049124192021002005>
- Cudeck & Browne (1983). Cross-validation of covariance structures, *Multivariate Behavioral Research*, 18(2), 147 – 167, DOI: 10.1207/s15327906mbr1802_2
- Hu, L.-T., & Bentler, P. M. (1998). Fit indices in covariance structure modeling: Sensitivity to underparameterized model misspecification. *Psychological Methods*, 3(4), 424-453. <https://doi.org/10.1037/1082-989X.3.4.424>
- McNeish, D., & Wolf, M. G. (2021). Dynamic fit index cutoffs for confirmatory factor analysis models. *Psychological Methods*. Advance online publication. <https://doi.org/10.1037/met0000425>
- Yuan, K.-H., Chan, W., Marcoulides, G. A., & Bentler, P. M. (2016). Assessing structural equation models by equivalence testing with adjusted fit indexes. *Structural Equation Modeling: A Multidisciplinary Journal*, 23(3), 319-330. <https://doi.org/10.1080/10705511.2015.1065414>